Muhammad Muizz Zafar

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Skills

Programming Languages:	Python, SQL, C#, Java, JavaScript, C++, HTML/CSS
Tools & Frameworks:	AWS (Lambda, S3, CloudWatch), Kubernetes (basic), Docker, Jenkins, CI/CD, Git

Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - Dec 2024

Work experience

Software Developer — May 2026 - Present
Elections Canada — Ottawa, ON

- Implemented a reusable ETL validation framework using **Python** and **PySpark** to validate and monitor **data pipelines**, improving data quality across multiple environments.
- Developed **.NET Core** REST APIs backed by SQL Server databases, enabling internal teams to securely access and consume operational data through reusable endpoints.

Software Developer Intern — September 2024 - April 2026
Kinaxis — Ottawa, ON

- Wrote **C++** code to address **incompatibilities** for Scope Controller. Also implemented unit test for each incompatibility to verify functionality
- Worked on the modernization of **KUnit** tests to **GTest** through serilization by writing **C++** code for capturing data configurations at Build, Publish, and Solve phases
- Automated the test generation process using **Python** to combine the **C++** files generated for the different phases and run queries against dataserver to verify the auto-generated output by the serilization
- Worked closely with dataserver team in order to move the **DevUnitTest** to algorothms build which helped for quicker build times and only relavent tests being run in the Jenkins pipelines
- Developed worksheets for **known differences** according to requirements by the PM in the maestro product
- Wrote queries to automate the detection of known difference between mocha and latte codebase. Also developed and maintained algotests for each known difference.
- Developed several **GTest** and **Algotests** for feature implemented in the Mocha codebase to check validity

Cloud Developer Intern — May 2023 - August 2023
Trend Micro Inc. — Ottawa, ON

- Developed serverless monitoring workflows using **AWS Lambda, S3, and CloudWatch** to analyze **50,000+ artifacts per day** and support automated vulnerability scanning pipelines for high-volume cloud artifacts.
- Implemented concurrent logic in **GoLang** using goroutines to improve throughput and strengthen error handling.
- Integrated vulnerability scanning tools (**Anchore, Grype**) with AWS services to automate security checks and operational monitoring.

Cloud Developer Intern — September 2022 - December 2022
Trend Micro Inc. — Ottawa, ON

- Built a serverless automation pipeline using **AWS Lambda, S3, and Athena** to process incoming cloud data events, query metadata, and trigger downstream updates, reducing manual operational effort.
- Configured **Splunk** and **AWS CloudWatch** monitoring, alarms, dashboards, and alerts to track **15+ services** and improve observability of production cloud infrastructure.
- Developed **Docker-based integration tests** and unit tests to improve reliability of distributed serverless data pipelines.

Software Developer Intern — September 2023 - April 2024
University of Ottawa — Ottawa, ON

- Redesigned the University of Ottawa's emergency alert website with **C# (.Net Core)** and **Angular**, introducing a modern, interactive UI. Used by over **20,000** students daily. website: ue.uottawa.ca/
- Optimized existing **C#** REST APIs by reducing **LINQ** query execution time for event data lookup by **30 %** and lines of code by **40 %**

Full Stack Developer Intern — January 2022 - April 2022
Applied Mind Inc. — Ottawa, ON

- Utilized containerization concepts to create **Docker** containers for the full- stack application for PROD deployment
- Automated the event detection process by adding events to **PostgreSQL** DB using **Python** and **GraphQl** which helped reduce testing time by **10 seconds**
- Added dynamic updates to the graph, map, and notification components using **Vue** dynamic update feature and **GraphQL** subscriptions which improved UX